# 3.14 Principal Ideal Domains

**Definition 3.14.1 (Principal Ideal Domain)**
*An integral domain $A$ is a **principal ideal domain** (or **PID**) if every ideal $\mathfrak{a}$ is principal.*

**Proposition 3.14.2**
*A PID is Noetherian.*

*Proof.* Suppose we have an ascending chain of ideals

$$\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots \subseteq \mathfrak{a}_n \subseteq \cdots$$

Clearly the union is again an ideal, which is also principal of the form $(a)$. We must have $a \in \mathfrak{a}_n$ for some $n$, whence it terminates after $n$. ∎

**Proposition 3.14.3 (Integers form a PID)**
*$\mathbb{Z}$ is a PID.*

*Proof.* This follows from the well-ordering principle. Let $\mathfrak{a}$ be an ideal with minimal positive element $d$. We claim $\mathfrak{a} = (d)$. By the division algorithm (or apply well-ordering principle to the coset $x + (d)$), for every $x \in \mathfrak{a}$ there is $0 \leq r < d$ and $q \in \mathbb{Z}$ such that

$$x = qd + r.$$

Clearly $r \in \mathfrak{a}$, whence by minimality $r = 0$ as required. ∎

**Proposition 3.14.4**
*Let $k$ be a field then the polynomial ring $k[X]$ is a PID*

*Proof.* Let $\mathfrak{a} \triangleleft k[X]$ be an ideal. Choose $f \in \mathfrak{a}$ to have minimal degree, then we claim $\mathfrak{a} = (f)$. For $g \in \mathfrak{a}$ we have by (3.7.14) $g = qf + r$ for $\deg(r) < \deg(f)$. Clearly $r \in \mathfrak{a}$, so by minimality $r = 0$ and the result follows. ∎

**Lemma 3.14.5 (Co-prime elements in a PID)**
*Let $A$ be a PID, then $x, y$ are coprime if and only if they have no non-invertible common divisors.*

*Proof.* First suppose $(x, y) = A$, then $ax + by = 1$ and any common divisor $d$ must divide 1 and therefore be invertible.

Conversely suppose $(x, y) \neq (1)$, since $A$ is a PID it must equal $(d)$ for some non-invertible $d$ which is then a common divisor. ∎

**Proposition 3.14.6**
*Let $A$ be a principal ideal domain, $M$ a free $A$-module with rank $m$ and $N \subset M$ an $A$-submodule. Then $N$ is free with rank $n \leq m$.*

*Proof.* The rank of a finite free $A$-module is well-defined (3.8.15).

We proceed by induction on $m$. If $m = 0$ then $M = 0$ and so $N = 0$ is free of rank 0. Suppose $m \geq 1$ and the result holds for rank $m - 1$. Let $x_1, ..., x_m$ be a basis for $M$ and let $\pi : M \to A$ be the map $\sum_i c_i x_i \mapsto c_m$. Its kernel is the free submodule $M'$ generated by $x_1, ..., x_{m-1}$, of rank $m - 1$. Then $N' := N \cap M'$ is the kernel of $\pi|_N$ and a submodule of $M'$, hence free of rank $n' \leq m - 1$ by the induction hypothesis. The image $\pi(N)$ is an ideal of $A$, hence equal to $(a)$ for some $a \in A$.

If $a = 0$ then $N = N'$ and we are done. Suppose $a \neq 0$ and choose $y \in N$ with $\pi(y) = a$. We claim $N = N' \oplus Ay$ with $Ay \cong A$. For $z \in N$ we have $\pi(z) = ca$ for some $c \in A$, whence $z - cy \in N'$. Moreover $\pi(cy) = ca$ is non-zero for $c \neq 0$, as $A$ is an integral domain. Hence $c \mapsto cy$ is injective, so $Ay \cong A$, and $N' \cap Ay = 0$ as $cy \in N'$ forces $ca = 0$ and so $c = 0$. Therefore $N$ is free with basis a basis of $N'$ together with $y$, of rank $n' + 1 \leq m$. ∎

**Proposition 3.14.7**
*Let $A$ be a principal ideal domain and $M$ a finitely generated, torsion-free $A$-module. Then $M$ is a finite free $A$-module.*

*Proof.* Let $\{m_1, ..., m_n\}$ be a generating set for $M$, and $\{v_1, ..., v_r\}$ an arbitrary linearly independent subset of $M$. Then by definition we have

$$v_i = \sum_{j=1}^{n} a_{ij} m_j \quad a_{ij} \in A$$

Consider the elements of $A^n$ given by

© David Rufino CC BY-NC-ND 4.0

$$w_i := (a_{i1}, ..., a_{in})$$

Then we can see that $\{w_1, ..., w_r\}$ is an $A$-linearly independent subset of $A^n$, and therefore a $K$-linearly independent subset of $K^n$ for $K = \mathrm{Frac}(A)$ (a $K$-linear dependence gives an $A$-linear dependence on clearing denominators). As $K^n$ has the standard basis of order $n$, it has dimension $n$ over $K$, so $r \leq n$ (3.3.126).

Therefore we may choose $\{v_1, ..., v_r\}$ to be a maximal linearly independent subset and define $M' := \langle v_1, ..., v_r \rangle$. For each generator $m_i$, by maximality of the subset there must exist an $a_i \neq 0$ such that

$$a_i m_i \in M'$$

Define $a := a_1 ... a_n$ then this satisfies $aM \subseteq M'$, since the $m_i$ generate $M$. As $A$ is an integral domain we have a series of injections

$$M \hookrightarrow aM \subseteq M' \xrightarrow{\sim} A^r$$

Therefore we are done by (3.14.6). ■

© David Rufino CC BY-NC-ND 4.0